

Hardness of finding large independent sets in three-colorable graphs

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Abstract

We prove that, for every fixed $0 < \delta < 1/3$, it is NP-hard to distinguish three-colorable graphs from graphs in which every independent set has fewer than δ times the number of vertices. Consequently, for every fixed integer $c \geq 3$, finding a proper c -coloring of a three-colorable graph is NP-hard.

1 Introduction

A proper coloring of a graph assigns different colors to adjacent vertices. An independent set is a set of pairwise nonadjacent vertices. For a nonempty finite graph G , write $n = |V(G)|$ and let $\alpha(G)$ be its maximum independent-set size. Every three-colorable graph contains an independent set of size at least $n/3$: one of its three color classes has that size. We prove that distinguishing this structured case from graphs with an arbitrarily small fixed independence ratio is NP-hard.

Theorem 1.1. *For every fixed real $0 < \delta < 1/3$, there is a deterministic polynomial-time algorithm R_δ that maps each explicitly encoded Boolean 3-CNF formula ϕ to a nonempty finite simple undirected unweighted graph G_ϕ such that*

$$\begin{aligned}\phi \text{ is satisfiable} &\implies G_\phi \text{ is three-colorable,} \\ \phi \text{ is unsatisfiable} &\implies \alpha(G_\phi) < \delta |V(G_\phi)|.\end{aligned}$$

The running time and explicit output size are polynomial in the ordinary binary encoding length of ϕ . Their constants and polynomial degree may depend on the fixed δ , but not on ϕ . The coloring in the first implication covers every vertex and is not supplied with the graph.

It suffices to prove Theorem 1.1 for rational δ . For a fixed real threshold, choose a fixed rational $0 < \delta_0 < \delta$ and use R_{δ_0} . We assume δ rational in the construction and its analysis. The density conclusion is stronger than an arbitrarily large fixed lower bound on chromatic number. Indeed, small independence ratio forces large chromatic number, whereas adjoining isolated vertices preserves chromatic number and can make the independence ratio arbitrarily close to one.

Approximate graph coloring asks, for fixed integers $3 \leq k \leq c$, to find a proper c -coloring of a graph promised to be k -colorable. Its decision version distinguishes k -colorable graphs from graphs that are not c -colorable. The following consequence recovers the constant-palette hardness theorem.

Corollary 1.2. *For every fixed pair of integers $3 \leq k \leq c$, it is NP-hard to distinguish k -colorable graphs from graphs that are not c -colorable.*

Proof. Choose a rational $0 < \delta < \min\{1/3, 1/c\}$ and apply Theorem 1.1. A three-colorable graph is k -colorable. A c -colorable graph has an independent set of size at least n/c , which the soundness bound excludes. $\square$

1.1 History and comparison with prior work

The difficulty of approximate coloring persists even with a very small promised palette. Khanna, Linial, and Safra established hardness of four-coloring three-colorable graphs [18]; Guruswami and Khanna later gave a different proof and obtained bounded-degree and edge-error variants [14]. The algebraic theory of promise constraint satisfaction developed by Barto, Bulín, Krokhin, and Opršal gave hardness of $(2k - 1)$ -coloring k -colorable graphs, and hence five colors when $k = 3$ [5, Theorem 6.5]. Their Example 2.9 records the conjectured hardness for every fixed $3 \leq k \leq c$.

Guruswami and Sandeep showed that the ordinary d -to-1 Games Conjecture with perfect completeness, for any fixed $d \geq 2$, implies constant-palette hardness for three-colorable graphs [15, Theorem 1]. Their reduction first gives arbitrarily small independent-set density with a $2d$ -colorable completeness promise [15, Theorem 9]. The arc-graph reduction of Krokhin, Opršal, Wrochna, and Živný [19, Theorem 1.8] then reduces the promised palette to three for the coloring conclusion. The resulting three-colorable hardness statement does not include the independent-set guarantee. The companion article [22, Theorem 1.1] establishes the ordinary perfect-completeness premise for $d = 2$.

Fei, Minzer, and Wang have proved the 4-to-1 Games Conjecture with perfect completeness [12, Theorem 1.6]. Combining it with the preceding reductions resolves the constant-palette conjecture: their Corollary 1.7 gives the three-colorable versus not c -colorable gap for every fixed c . Their Corollary 1.9 gives arbitrarily small independent-set density with an eight-colorable completeness promise, corresponding to $2d = 8$. Theorem 1.1 combines this stronger form of soundness with three-colorability of the entire graph. Our proof starts with ordinary projection Label Cover and does not require a bound on projection-fiber sizes.

The distinction between colorability and independence density also appears in earlier conditional results. Dinur, Mossel, and Regev obtained three-colorable completeness and arbitrarily small independent-set soundness from their fish-shaped Label Cover conjecture [10, Conjecture 4.8 and Sections 4.3–4.5]. Braverman, Khot, Lifshitz, and Minzer obtained the same graph promise from the Rich 2-to-1 Games Conjecture using their invariance principle for the multi-slice [6, Corollary 1.19]. The Rich condition requires that, at each left question, a uniformly random incident constraint induce a uniformly random partition of the left alphabet among all partitions into pairs. This is not part of the ordinary 2-to-1 premise above.

A different line of work relaxes completeness by permitting deletion of a small fraction of vertices. Dinur, Khot, Perkins, and Safra proved that, for arbitrarily small fixed $\varepsilon > 0$, it is NP-hard to find an independent set of density $1/9$ when an induced three-colorable subgraph occupies at least $(1 - \varepsilon)n$ vertices [9]. Hecht, Minzer, and Safra subsequently proved, under randomized reductions, hardness of almost coloring such almost-three-colorable graphs with any fixed palette [16]. The all-vertex completeness in Theorem 1.1 retains the original coloring promise, while allowing every fixed positive soundness density.

Algorithmic work gives a complementary perspective on the remaining quantitative gap. Bansal, Huang, and Lee give a polynomial-time algorithm using $O(n^{0.19539})$ colors [4], improving the $\tilde{O}(n^{0.19747})$ bound of Kawarabayashi, Thorup, and Yoneda [17]. In preprints posted after the September 24 version of this manuscript, Narang and Tang report a randomized polynomial-time bound of $O(n^{(13 - \sqrt{97})/18 + \varepsilon})$ for each fixed $\varepsilon > 0$ [21], and Anand reports a randomized polynomial-time bound of $O(n^{4/23})$ [1]. Our reduction has fixed δ and fixed palettes; its polynomial exponent can depend on these constants.

1.2 Proof overview and method ancestry

The starting problem is Label Cover: a bipartite collection of questions and tests, each prescribing a projection from a left answer to a right answer. The PCP theorem and parallel repetition supply instances in which either every test can be satisfied or every labeling succeeds on at most an arbitrarily small fixed fraction of tests [2, 24, 11].

We replace one left question at a time by a right question, obtaining chains of question tuples. This use of layered tuples and projection constraints is related to the multilayered PCP construction of Dinur, Guruswami, Khot, and Regev [7]. At each tuple, a point assigns a phase in $\mathbb{T} = \mathbb{R}/\mathbb{Z}$ to every possible product answer. An answer projection $\pi : M \rightarrow M'$ pulls a phase vector $x \in \mathbb{T}^{M'}$ back to $x \circ \pi \in \mathbb{T}^M$. Link these two points with length zero, and link points within each phase space with their sup circle distance. Shortest paths through these links define a pseudometric. Edges compare one point with the coordinatewise half-shift of another. A satisfying Label Cover labeling evaluates each phase assignment at one product answer. This evaluation is nonexpanding, and the edge threshold forces adjacent vertices to have circle phases more than $1/3$ apart. Three equal half-open intervals of the circle therefore color all vertices properly.

For soundness, an independent set gives bounded functions on the tuple phase spaces that change sign under a half-shift and satisfy a common Lipschitz bound. Austin’s dimension-independent approximation theorem [3], a continuous counterpart of Friedgut’s junta theorem [13], approximates these functions using boundedly many answer coordinates. The difficulty is that approximation under product measure need not survive identifications of coordinates by projections. We therefore choose coordinate lists for whole small subchains and introduce independent rotation variables at every layer of each subchain. All required compatibility properties are proved within this paper. Projecting the selected coordinates to the last layer places all lists in a common answer set. A shared answer for two separated subchains supplies compatible answers to an intervening Label Cover test, so low source value makes these intersections rare.

Two further steps turn this structural information into a density bound. First, a distributional alignment lemma chooses a law on sets of layers before seeing the lists. On most selected sets, each listed label can be assigned to an index shared by all its occurrences. The bound is independent of the label universe. The proof combines finite minimax, Ramsey homogenization of finite probability patterns, and an order-invariant random-array argument. The homogenization step has antecedents in the Ramsey theory of random variables [25]; we give the particular alignment argument in full.

Second, we sample coefficients at finitely many levels between 0 and 1, with geometrically decreasing probabilities. Enough layers make a coefficient equal to 1 likely. That layer absorbs auxiliary uniform phase shifts without changing their distribution. Alignment makes dependence on a zero-coefficient layer small; decreasing a positive coefficient by one level bounds its probability of substantial dependence. A dimension-independent bound on influential coordinates then gives a small mean square, and hence a small independent-set density.

Only fixed finite rational data enter the graph construction. The probability experiment is expanded exactly into unweighted vertices, so the reduction is deterministic. Section 2 states the standard inputs and derives the analytic coordinate bounds. Section 3 constructs the graph with finite parameters and proves completeness. Section 4 extracts the coordinate lists from an independent set and decodes intersections of separated lists. Section 5 states distributional alignment and applies it to these lists. Section 6 proves the coefficient estimate under explicit inequalities, and Section 7 chooses all constants in dependency order and completes Theorem 1.1. The self-contained proof of distributional alignment is given in Section 8.

2 Two standard inputs and an analytic consequence

We first state the two established results used in the reduction. The new combinatorial and analytic arguments will be proved in full. For a positive integer r , write $[r] = \{1, \dots, r\}$. All products of metric spaces carry the sup metric unless stated otherwise. The circle $\mathbb{T} = \mathbb{R}/\mathbb{Z}$ carries the distance $d_{\mathbb{T}}(x, y) = \min_{k \in \mathbb{Z}} |x - y - k|$. Uniform inputs on a cube have product Lebesgue law, and uniform inputs on a product of circles have product Haar law. Norms $\|\cdot\|_p$ always use the indicated probability measure. An empty sum is zero.

2.1 A projection constraint problem

A *Label Cover instance* consists of finite question sets U, V , nonempty finite answer alphabets Σ_L, Σ_R , and a nonempty explicit list $\mathcal{C}$ of test occurrences. An occurrence c has questions $u_c \in U$, $v_c \in V$, and a total map $\pi_c : \Sigma_L \rightarrow \Sigma_R$. The value is

$$\text{val}(\mathcal{C}) = \max_{\ell: U \rightarrow \Sigma_L, \rho: V \rightarrow \Sigma_R} \Pr_{c \in \mathcal{C}} [\pi_c(\ell(u_c)) = \rho(v_c)],$$

where occurrences, including repeated occurrences, are sampled uniformly.

Theorem 2.1 (Perfect-completeness Label Cover). *For every fixed rational $\sigma \in (0, 1)$ there are nonempty constant alphabets and a deterministic polynomial-time reduction from 3SAT to explicit Label Cover instances with nonempty occurrence lists such that satisfiable formulas have value 1 and unsatisfiable formulas have value at most σ .*

This is the standard consequence of the PCP Theorem and parallel repetition [2, 24]. A quantitative statement is [11, Theorem 6.2, full version]: for a fixed number k of repetitions, the output size is $n^{O(k)}$, the alphabet size is at most a^k , and the soundness is at most β_0^k , for absolute constants $a > 1$ and $0 < \beta_0 < 1$. Choose a fixed k for which $\beta_0^k \leq \sigma$. The unweighted total-projection formulation is explicitly recorded in [8, Definitions 1.1–1.2 and Theorem 1.3]. The latter source even supplies d -to-one maps, a property we do not use. Here the verifier’s random choices specify an explicit list of constraints; they are not random choices of the reduction. Unused questions can be removed.

2.2 Juntas for the sup metric

A function is a *junta on S* if it depends only on the input coordinates in S . The approximating functions below need not be continuous. We use Austin’s continuous junta theorem [3], which extends the dimension-independent coordinate approximation phenomenon of Friedgut’s Boolean junta theorem [13].

Theorem 2.2 (Dimension-independent junta approximation). *For every $0 \leq L < \infty$ and $\tau > 0$ there is an integer $J(L, \tau) \geq 1$ such that every L -Lipschitz function $f : [0, 1]^N \rightarrow [-1, 1]$, for every positive integer N , admits a junta on at most $J(L, \tau)$ coordinates with L^2 error less than τ . The junta may be taken to be a coordinate conditional expectation of f .*

To obtain this formulation from [3, Theorem 1.1, arXiv version 4], use the usual interval with uniform measure and modulus of continuity $\omega(t) = Lt$. The interval is compact, connected, and locally connected, as required there. Austin’s theorem gives $\|f - \mathbb{E}[f \mid X_S]\|_1 < \tau^2/2$ with $|S|$ bounded independently of N . Both functions take values in $[-1, 1]$, so

$$\|f - \mathbb{E}[f \mid X_S]\|_2^2 \leq 2\|f - \mathbb{E}[f \mid X_S]\|_1 < \tau^2.$$

The bound applies to pullbacks of torus functions, since the quotient $[0, 1]^N \rightarrow \mathbb{T}^N$ is nonexpanding and preserves the product uniform law.

For a square-integrable function H of independent circle coordinates $z_1, \dots, z_s$, let E_i average coordinate z_i alone, and set

$$Q_i = \text{id} - E_i, \quad D_i(H) = \|Q_i H\|_2.$$

Both E_i and Q_i are orthogonal projections. In particular, $\|Q_i f\|_2 \leq \|f\|_2$; the constant here is one.

Lemma 2.3 (Two dimension-independent bounds). *Fix $0 \leq L < \infty$ and $\varepsilon > 0$. There are $\beta > 0$ and an integer $K \geq 1$, independent of s , such that every L -Lipschitz $H : \mathbb{T}^s \rightarrow [-1, 1]$ satisfies:*

- (i) *if $\mathbb{E}H = 0$ and $\mathbb{E}H^2 > \varepsilon$, then $D_i(H) \geq \beta$ for some i ;*
- (ii) *at most K indices satisfy $D_i(H) \geq \beta/2$.*

Proof. Let $J = J(L, \sqrt{\varepsilon}/2)$ from Theorem 2.2. Choose a set S of at most J coordinates such that $G = \mathbb{E}[H \mid z_S]$ satisfies $\|H - G\|_2 < \sqrt{\varepsilon}/2$. If H has mean zero and squared norm exceeding ε , then G has mean zero and, by orthogonality, $\|G\|_2^2 > 3\varepsilon/4$, in particular $\|G\|_2^2 > \varepsilon/4$.

For completeness, decompose

$$H = \sum_{A \subseteq [s]} H_A, \quad H_A = \left(\prod_{i \in A} Q_i \right) \left(\prod_{i \notin A} E_i \right) H.$$

The factors commute and the summands are orthogonal. Thus $G = \sum_{A \subseteq S} H_A$ and, in the mean-zero case,

$$\|G\|_2^2 = \sum_{\emptyset \neq A \subseteq S} \|H_A\|_2^2 \leq \sum_{i \in S} \sum_{A \ni i} \|H_A\|_2^2 = \sum_{i \in S} D_i(H)^2.$$

Consequently $\beta = \sqrt{\varepsilon/(4J)}$ works in (i).

For (ii), use Theorem 2.2 again with error less than $\beta/2$, and write G' for the resulting junta, on at most $K = J(L, \beta/2)$ coordinates. If i is not one of these coordinates, then $Q_i G' = 0$, so

$$D_i(H) = \|Q_i(H - G')\|_2 \leq \|H - G'\|_2 < \beta/2.$$

This proves the cardinality bound. □

The first part locates a coordinate when the function has substantial variance; the second limits how many coordinates can have a smaller but still fixed amount of dependence. Their simultaneous dimension independence is what will permit the number of test blocks to be chosen after both bounds are fixed.

3 The finite graph reduction

We first construct a graph from an arbitrary Label Cover instance and fixed finite parameters. A satisfying labeling will give its three-coloring. The following sections analyze independent sets and determine which parameter choices make their density small.

Fix integers $r \geq s \geq 2$ and $m \geq 1$, an even positive integer P , a rational $\lambda \in (0, 1)$, and a rational probability distribution μ on $\binom{[r]}{s}$. Set $D = mP$. The coefficient values and their probabilities are

$$\mathcal{T}_m = \{0, 1/m, \dots, 1\}, \quad p_k = \Pr(t = k/m) = \frac{\lambda^k}{\sum_{\ell=0}^m \lambda^\ell} \quad (0 \leq k \leq m). \quad (1)$$

These finite data are inputs to the construction. Section 7 will choose them as constants depending only on the target density δ , before choosing the Label Cover soundness and alphabets. The finite probability experiment below specifies a deterministic list of vertices by expanding each rational probability into a multiplicity.

Choose a fixed integer $q > 1/\delta$. Before invoking Label Cover, reindex the occurring variables densely, preserving repeated occurrences of each variable. Remove tautological clauses and duplicate literals. A formula with no clauses maps directly to a one-vertex graph; a formula containing an empty clause maps directly to K_q . Pad a remaining short clause to width three by including all sign choices for fresh filler variables. This preserves satisfiability at constant overhead. In the remainder, the formula has only nonempty clauses and has passed through this preprocessing.

3.1 Tuples, grids, and zero-length links

Let $\mathcal{C}$ be any Label Cover instance, with unused questions removed. In layer $i \in [r]$, take all question tuples

$$\mathbf{q} = (v_1, \dots, v_{i-1}, u_i, \dots, u_{r-1}) \in V^{i-1} \times U^{r-i}.$$

The answer set of such a tuple is

$$M_i = \Sigma_R^{i-1} \times \Sigma_L^{r-i}.$$

Layered products of questions with projection constraints also occur in the multilayered PCP construction of [7]. Here each position is a separate coordinate even if question names repeat. Different tuples in a layer use separate copies of this same answer set.

Between adjacent layers i and $i+1$, impose a map whenever the tuples agree outside position i and their questions at position i are the endpoints of a test occurrence c . The map on answer sets applies π_c in position i and the identity in every other position. Impose it separately for every occurrence.

Write $\Gamma = (D^{-1}\mathbb{Z})/\mathbb{Z}$. At each question tuple place a separate copy of Γ^{M_i} , and let $\mathcal{X}$ be their disjoint union. Make a finite undirected weighted link graph on $\mathcal{X}$:

- Within each copy, link any two points with length their sup circle distance.
- For every imposed map $\pi : M_i \rightarrow M_{i+1}$, link each target point $x \in \Gamma^{M_{i+1}}$ to its source pullback $x \circ \pi$ with length zero.

Let dist be the resulting shortest-path extended pseudometric, with distance infinity between components. Thus distinct points may have distance zero.

Let T add $1/2$ to every phase coordinate, within each copy. Since D is even, T permutes $\mathcal{X}$ and satisfies $T^2 = \text{id}$. It preserves all link lengths and therefore is an isometry of dist .

If some $x \in \mathcal{X}$ satisfies

$$\text{dist}(x, Tx) \leq 1/8, \tag{2}$$

output K_q and stop. This branch already has independent-set density $1/q < \delta$. We will show that it never occurs in completeness.

3.2 Vertices and edges

If (2) fails for every point, consider the following finite experiment.

1. Sample $r-1$ independent uniform occurrences $c_1, \dots, c_{r-1}$, with questions (u_h, v_h) at position h . They determine a chain of tuples

$$\mathbf{q}_i = (v_1, \dots, v_{i-1}, u_i, \dots, u_{r-1}), \quad i \in [r].$$

Write $\pi_{ij} : M_i \rightarrow M_j$ for the corresponding composite answer maps. They apply the test projections in positions $i, \dots, j-1$ and leave the other positions unchanged; in particular they are composition-consistent.

2. Independently choose $B \sim \mu$. For every $j \in B$, independently choose t_j from (1), and choose independent rotation entries

$$\hat{\theta}_{j,\ell} \text{ uniform in } \{0, 1/P, \dots, (P-1)/P\}, \quad \ell \in M_j.$$

3. With $b = \min B$, give the outcome the location

$$x = \left(\sum_{j \in B} t_j \hat{\theta}_{j, \pi_{bj}(\kappa)} \pmod{1} \right)_{\kappa \in M_b} \in \Gamma^{M_b} \quad (3)$$

in the copy at tuple $\mathbf{q}_b$.

Take distinct output vertices for these elementary outcomes with integer multiplicities that make the experiment their exact uniform law. The existence and size of these multiplicities are verified below. Different vertices may have the same location. For distinct vertices with locations x, y , put an edge precisely when

$$\text{dist}(x, Ty) \leq 1/8. \quad (4)$$

This is symmetric because T is an involutive isometry: $\text{dist}(x, Ty) = \text{dist}(Tx, y) = \text{dist}(y, Tx)$.

Proposition 3.1 (Explicit deterministic encoding). *For fixed choices of the parameters and Label Cover alphabets, this rule constructs a nonempty finite simple unweighted graph in deterministic polynomial time and output size, measured in the size of the explicit Label Cover instance. Composing with Theorem 2.1 for any fixed σ gives polynomial time and output size in the binary encoding length of the original 3-CNF formula.*

Proof. Let $N = |\mathcal{C}| \geq 1$. All alphabets, layer counts, and grids are fixed constants. There are polynomially many question tuples, each with a constant-size grid. The imposed maps and links are therefore polynomially enumerable. Every finite link length is an integer multiple of $1/D$. After multiplying lengths by D , exact integer shortest-path algorithms compute all distances, with infinity stored separately. Shortest paths have polynomial bit length, and comparison with $1/8$ is exact by multiplication by 8. This also implements the clique test.

There are exactly N^{r-1} equally likely occurrence chains. For each chain, the remaining outcome probabilities are fixed rational numbers. The number of rotation entries can vary with B and its layers, but it ranges over fixed constants. Choose a common positive denominator C_0 for all these conditional probabilities. For a conditional outcome of probability p , emit $C_0 p$ distinct vertices, omitting zero-probability outcomes. Every chain then contributes exactly C_0 vertices. The total is $C_0 N^{r-1}$, and uniform sampling of these vertices is exactly the stated experiment, followed by a uniform choice among the copies of the chosen outcome. Exhaustive enumeration uses no random bits. Evaluating (4) for every distinct pair remains polynomial and gives an explicit edge list with no loops or multiple edges.

The Label Cover reduction is polynomial for the fixed σ . The initial variable reindexing and clause preprocessing take polynomial time in the ordinary binary input length; the two trivial branches have constant-size outputs. Thus the construction covers every explicitly encoded 3-CNF formula. $\square$

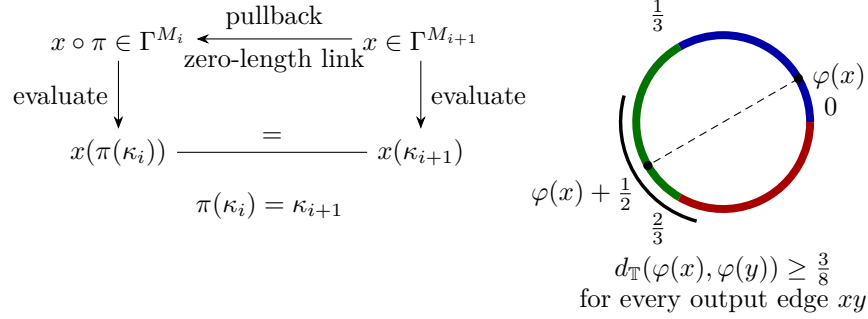


Figure 1: The auxiliary geometry and the output coloring. Left: a satisfying product answer makes evaluation agree across a projection link. Here κ_i and κ_{i+1} are the satisfying product answers at the two tuples. The answer map $\pi : M_i \rightarrow M_{i+1}$ and its phase pullback go in opposite directions. Right: the three arcs represent the half-open color intervals. For the illustrated phase $\varphi(x)$, the black outer arc contains the possible phases $\varphi(y)$ of its neighbors, within $1/8$ of the antipodal phase. The weighted links on the left define the pseudometric; the output graph uses the separate half-shift edge rule.

Lemma 3.2 (All-vertex completeness). *If the Label Cover instance has value 1, the clique branch does not occur and the constructed graph has a proper three-coloring.*

Proof. Fix a labeling satisfying every occurrence. At each question tuple, form the product of its assigned answers, an element of that tuple’s answer set. Evaluate a grid point at this product answer. This defines a map $\varphi : \mathcal{X} \rightarrow \mathbb{T}$.

Within a copy, evaluation is nonexpanding for the sup circle distance. Across each zero link the two evaluations agree because the labeling satisfies that test occurrence. Hence φ is nonexpanding for every path, and thus for dist. Also $\varphi(Tx) = \varphi(x) + 1/2$. Consequently $\text{dist}(x, Tx) \geq 1/2$ for every x , so the clique test fails.

If output vertices at x, y are adjacent, then

$$d_{\mathbb{T}}(\varphi(x), \varphi(y) + 1/2) \leq 1/8, \quad d_{\mathbb{T}}(\varphi(x), \varphi(y)) \geq 3/8 > 1/3.$$

Color each vertex according to which of the half-open intervals $[0, 1/3)$, $[1/3, 2/3)$, $[2/3, 1)$ contains its phase. Two phases in the same interval have circle distance less than $1/3$, so this colors every vertex properly. All multiplicity copies receive the color of their location. $\square$

Figure 1 shows how the two parts of the construction serve this coloring: projection links preserve the evaluated phase, whereas an output edge separates its endpoint phases by more than the length of a color interval.

4 From an independent set to bounded subchain lists

Fix $\sigma \in (0, 1)$ and assume that the Label Cover value is at most σ . The clique branch already has the required soundness, so suppose it does not occur. Fix a nonempty independent vertex set $\mathcal{A}$ in the output graph. We first turn it into functions on the tuple tori, then approximate their restrictions to small subchains using bounded coordinate lists. Source soundness will show that lists belonging to separated subchains are usually disjoint. Throughout the analysis, set $L = 64$.

4.1 Odd functions and compatibility

Let $A \subseteq \mathcal{X}$ be the set of locations of vertices in $\mathcal{A}$, with multiplicities removed. Then

$$\text{dist}(A, TA) > 1/8. \quad (5)$$

For locations of distinct selected vertices this follows from the missing edge in (4). For a point compared with itself, it follows from exclusion of (2). Since the sets are finite, these pointwise strict inequalities give (5), including when distances are infinite.

On $\mathcal{X}$ define

$$f(x) = (1 - 32 \text{dist}(x, A))_+ - (1 - 32 \text{dist}(Tx, A))_+, \quad (u)_+ = \max\{u, 0\}, \quad (6)$$

with each bump defined to be zero at infinite distance. This function takes values in $[-1, 1]$, satisfies $f(Tx) = -f(x)$, and equals 1 on A . It is constant across every zero link. Within each grid copy it is 64-Lipschitz for the sup circle metric: distance to A is 1-Lipschitz on a component meeting A , and its bump is identically zero on a component not meeting A . Also the shortest-path pseudometric never exceeds a within-copy link distance.

At each tuple $\mathbf{q}$, extend its grid function to the whole torus $\mathbb{T}^{M_i}$. Using the real-valued Lipschitz extension formula of McShane [20], set

$$g_{\mathbf{q}}(x) = \min_{y \in \Gamma^{M_i}} (f(y) + L d_{\infty}(x, y)),$$

where y ranges over that tuple's copy and d_{∞} is the sup circle metric. This is L -Lipschitz and equals f on the grid. Clip its values to $[-1, 1]$, obtaining $\tilde{g}_{\mathbf{q}}$, and put

$$F_{\mathbf{q}}(x) = \frac{\tilde{g}_{\mathbf{q}}(x) - \tilde{g}_{\mathbf{q}}(Tx)}{2}.$$

The resulting function is still an extension of the odd grid data, is L -Lipschitz, is valued in $[-1, 1]$, and is odd under T . Fix this entire family of functions before sampling any test chain. Along a chain write $F_i = F_{\mathbf{q}_i}$.

For every chain and every $i \leq j$, these extensions satisfy

$$|F_j(x) - F_i(x \circ \pi_{ij})| \leq 2L/D, \quad x \in \mathbb{T}^{M_j}. \quad (7)$$

Indeed, round each coordinate of x once to a grid point y within $1/D$. Pullback does not enlarge this error. The grid points y and $y \circ \pi_{ij}$ have equal f values by the chain of zero links, so the two extension errors total at most $2L/D$. Intermediate layers contribute no additional approximation error.

4.2 Bounded lists for a subchain

The functions F_i now supply the analytic data we need along any chain. For each small set of layers, we will choose a bounded list of answer coordinates that approximates every allowed coefficient choice. The bound must be independent of the Label Cover alphabets, and the choice must depend only on the data of that subchain.

Fix an approximation tolerance $\gamma > 0$. For $\emptyset \neq I \subseteq [r]$ with $|I| \leq s$, put $b_I = \min I$. Given $t \in \mathcal{T}_m^I$, define

$$h_{I,t}(\theta) = F_{b_I} \left(\left(\sum_{j \in I} t_j \theta_{j, \pi_{b_I j}(\kappa)} \pmod{1} \right)_{\kappa \in M_{b_I}} \right), \quad \theta \in \prod_{j \in I} [0, 1]^{M_j}. \quad (8)$$

The entries of θ are independent uniform real variables. This evaluates F_{b_I} at the continuous version of the sampled location in (3), using only the layers of I . The phase map has Lipschitz constant at most $|I| \leq s$: coordinate duplication by a projection cannot enlarge a sup distance. Thus $h_{I,t}$ is sL -Lipschitz, independently of the label-set sizes and projection fibers.

Lemma 4.1 (Bounded subchain lists). *For every fixed occurrence chain and every nonempty $I \subseteq [r]$ with $|I| \leq s$, there are sets $S_I^j \subseteq M_j$, $j \in I$, with*

$$\sum_{j \in I} |S_I^j| \leq d, \quad d := \max\{1, J(sL, \gamma)(m+1)^s\}, \quad (9)$$

such that, for every $t \in \mathcal{T}_m^I$, a junta $g_{I,t}$ using only scalar coordinates (j, ℓ) with $\ell \in S_I^j$ satisfies

$$\|h_{I,t} - g_{I,t}\|_2 < \gamma. \quad (10)$$

The lists can be chosen as functions only of the subchain's label sets, internal projection maps, and functions F_j , $j \in I$. Identical subchain data receive identical choices.

Proof. For each of the at most $(m+1)^s$ coefficient choices, apply Theorem 2.2 to $h_{I,t}$ with error γ . This uses at most $J(sL, \gamma)$ scalar coordinates. For each block j , take the union of its selected coordinates over all coefficient choices to obtain S_I^j and (9).

To make the choice depend only on the stated subchain data, order the scalar coordinates and choose the first subset of size at most $J(sL, \gamma)$ whose coordinate conditional expectation has error less than γ . The theorem guarantees such a subset. These choices enter only the soundness proof and are never computed by the reduction. In particular, a list includes approximating coordinates for every coefficient vector before any coefficients are sampled. $\square$

4.3 Why the lists can be decoded locally

Use the fixed rule of Lemma 4.1 along a uniform occurrence chain. Using $M_r = \Sigma_R^{r-1}$ as a common label universe, set

$$A_I = \bigcup_{j \in I} \pi_{j,r}(S_I^j), \quad \emptyset \neq I \subseteq [r], \quad |I| \leq s. \quad (11)$$

By (9), each A_I has size at most d . We first control intersections between lists on separated sets of layers.

Lemma 4.2 (Separated lists). *For every fixed I, J with $1 \leq |I|, |J| \leq s$ and $\max I < \min J$,*

$$\Pr_{\text{chain}}(A_I \cap A_J \neq \emptyset) \leq d^2 \sigma.$$

Proof. Put $h = \max I < \min J$ and condition on every occurrence except c_h . The remaining occurrence is still uniform in the original instance. At each layer in I , position h of the tuple is the left question u_h . Every internal projection between layers of I uses only tests with index less than h . Thus all subchain data for I , including its already fixed tuple functions, depend only on u_h and the conditioned background, not on the opposite question or on the projection of c_h . Likewise, all data for J depend only on v_h and the background. Figure 2 illustrates this separation.

Form a list of candidate left answers by taking position h of every label in every S_I^j for $j \in I$. This gives at most d elements of Σ_L , as a function of u_h and fixed background. The analogous list for J gives at most d elements of Σ_R as a function of v_h . Choose deterministic orderings and pad each list to length d with an arbitrary answer, also when the list is empty.

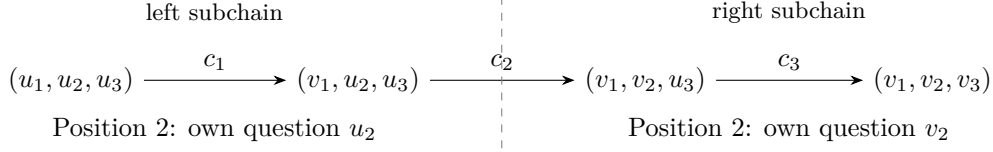


Figure 2: Four layers with the separating test c_2 left unconditioned. Once c_1, c_3 are fixed, the left and right subchain data depend only on the respective own question of c_2 . Answer maps act in the position indicated by each test.

If A_I and A_J intersect, some left and right selected labels have the same image at layer r . Equality in position h says that π_{c_h} sends the associated left candidate to the right candidate: on the left, exactly the test c_h acts in that position, and on the right the position is already a right-answer coordinate. For each pair of list positions, selecting those answers for each own question is a deterministic strategy for the original Label Cover instance. Its success probability over the uniform c_h is at most σ . A union bound over the d^2 position pairs proves the conditional bound, and averaging over the background proves the lemma. $\square$

The fixed functions may depend on the entire instance and on $\mathcal{A}$. This causes no loss of locality: they are fixed before the sampling experiment, so evaluating the family at a tuple reveals only that tuple. In particular, if two occurrences have the same own question but different opposite questions or projections, the list rule at that side uses identical data.

We have obtained bounded lists in a common answer set, and source soundness controls intersections of lists on separated layer sets. The next section turns this separation into a stronger property: each listed answer can be assigned to one layer common to all of its occurrences inside the sampled set B .

5 Aligning the lists on a sampled set of layers

The terminal lists $A_I \subseteq M_r$ all have size at most d . For the analytic argument, we need more than disjointness on separated sets: each label should have a layer common to every set on whose list it appears. These properties differ even for one label. A label that occurs on the three sets $\{1, 2\}$, $\{1, 3\}$, and $\{2, 3\}$ has no common index, although none of these sets is separated from another.

The following abstract lemma supplies the common-index property on a sampled s -element set. Its essential quantifier is that the sampling law is chosen before the lists, with no dependence on the label universe. This is what allows the same graph construction to handle every independent set.

Lemma 5.1 (Distributional alignment). *For every pair of integers $s, d \geq 1$ and every real $\eta > 0$, there exist an integer $r \geq s$ and a rational probability distribution μ on $\binom{[r]}{s}$ with the following property. Let M_* be any set, and let $A_I \subseteq M_*$, $|A_I| \leq d$, be specified for every nonempty $I \subseteq [r]$ with $|I| \leq s$. Assume that*

$$A_I \cap A_J = \emptyset \quad \text{whenever } \max I < \min J. \quad (12)$$

Then, with probability at least $1 - \eta$ over $B \sim \mu$, there exists a map $a : M_ \rightarrow B$ such that*

$$a(A_I) \subseteq I \quad \text{for every nonempty } I \subseteq B. \quad (13)$$

The map a may depend on the list family and on B .

The complete proof is in Section 8. It replaces labels by their finite equality patterns, applies minimax and Ramsey homogenization, and obtains an order-invariant random pattern. In that

limit, separated disjointness forces every label to have an index common to all of its occurrences. We now apply the lemma to the lists already extracted from an independent set.

Fix $\varepsilon > 0$, and suppose that the layer count r and law μ in the graph construction are supplied by Lemma 5.1 for $s, d, \eta = \varepsilon$. Section 7 will choose these data after the alphabet-independent bound d is fixed and before choosing the Label Cover instance.

Definition 5.2. A pair consisting of a chain and a set $B \in \binom{[r]}{s}$ is *aligned* if there is a map $a : M_r \rightarrow B$ such that $a(A_I) \subseteq I$ for every nonempty $I \subseteq B$.

Lemma 5.3 (Aligned pairs occur with high probability). *Under these choices, for independent uniform chain sampling and $B \sim \mu$, the probability that the pair is not aligned is at most $4^r d^2 \sigma + \varepsilon$.*

Proof. There are at most 4^r ordered pairs of subsets of $[r]$. By Lemma 4.2 and a union bound, the probability that some separated pair of terminal lists intersects is at most $4^r d^2 \sigma$. For each chain with no such intersection, Lemma 5.1 gives an alignment except for a set of B of μ -probability at most ε . Adding the two failure probabilities proves the bound. $\square$

The lists were selected before B , and include the juntas for every coefficient choice. An alignment for a fixed aligned pair can therefore be chosen before the actual coefficient and rotation draws. The next section uses this fixed alignment to control dependence on one auxiliary phase for each layer in B .

6 The coefficient test and soundness

The remaining task is to bound the density of an independent set after fixing an aligned pair of a chain and a selected set of layers. We add one auxiliary phase for each selected layer. Alignment makes the influence of a zero-coefficient layer small, whereas the geometric coefficient distribution makes large influences at positive coefficients unlikely.

Keep the tolerance $\varepsilon > 0$ of Section 5, and take $\beta > 0$ and $K \geq 1$ from Lemma 2.3 with $L = 64$ and this ε . For the graph parameters of Section 3 and the list-approximation tolerance γ of Section 4, assume

$$\begin{aligned} L/m < \beta/2, \quad \lambda K < \varepsilon, \quad (1 - p_m)^s < \varepsilon, \\ \frac{s(2\gamma)^2}{\beta^2} < \varepsilon, \quad \frac{2L}{D} < \gamma, \quad \frac{2Ls}{P} < \varepsilon. \end{aligned} \tag{14}$$

Section 7 will choose constants satisfying these inequalities. A coefficient equal to 1 is called a *full coefficient*. Although its individual probability p_m may be small, the third inequality makes at least one of the s coefficients full with probability greater than $1 - \varepsilon$.

Proposition 6.1. *Assume (14). Suppose that the construction does not take the clique branch, and let $\mathcal{A}$ be a nonempty independent set of its output graph. Form the functions and lists associated with $\mathcal{A}$ as above. For every chain and s -element set B admitting a map*

$$a : M_r \longrightarrow B, \quad a(A_I) \subseteq I \quad \text{for every nonempty } I \subseteq B,$$

the conditional probability that the sampled output vertex belongs to $\mathcal{A}$ is at most 6ε .

Proof. Fix such a chain, B , and a map a , before drawing any coefficients or rotations. Write $b = \min B$. Since $f = 1$ at every location of $\mathcal{A}$, it suffices to bound the expected square of F_b at

the sampled grid location by 6ε . We establish this through a continuous rotation experiment with auxiliary phases. Initially replace the discrete rotations by independent uniform real variables

$$\theta_{j,\ell} \in [0, 1), \quad j \in B, \quad \ell \in M_j.$$

All these variables are independent of the coefficient vector $t \in \mathcal{T}_m^B$. Introduce additional independent Haar-uniform variables $z = (z_i)_{i \in B} \in \mathbb{T}^B$, and define

$$H(t, \theta; z) = F_b \left(\left(z_{a(\pi_{br}(\kappa))} + \sum_{j \in B} t_j \theta_{j, \pi_{bj}(\kappa)} \pmod{1} \right)_{\kappa \in M_b} \right). \quad (15)$$

For fixed t, θ , the map from z to the phase vector in (15) is nonexpanding for the sup circle metrics: each output coordinate uses just one coordinate of z . Consequently H is L -Lipschitz as a function of z , independently of s and of all label cardinalities. It is bounded by one in absolute value. A simultaneous half-shift of the z coordinates half-shifts every input coordinate of F_b , so oddness gives

$$\mathbb{E}_z H(t, \theta; z) = 0. \quad (16)$$

Let E_i average only z_i and let $Q_i = \text{id} - E_i$, an orthogonal projection of norm one. We use the notation

$$D_i(t, \theta) = \|Q_i H(t, \theta; \cdot)\|_{L^2(\mathbb{T}^B)}, \quad X(t, \theta) = \mathbb{E}_z H(t, \theta; z)^2.$$

By Lemma 2.3 and (16),

$$X(t, \theta) > \varepsilon \implies \max_{i \in B} D_i(t, \theta) \geq \beta, \quad (17)$$

$$\#\{i \in B : D_i(t, \theta) \geq \beta/2\} \leq K \quad \text{for every } t, \theta. \quad (18)$$

Zero coefficients. Let

$$\mathcal{F} = \{t : \text{some } t_j = 1\}$$

be the event that a full coefficient is present. Fix $t \in \mathcal{F}$ and an index $i \in B$ with $t_i = 0$. Choose $j \in B$ with $t_j = 1$; necessarily $j \neq i$. Put $I = B \setminus \{i\}$ and $c = \min I$. The set I is nonempty. In the subchain function $h_{I, t|_I}$ from (8), leave all rotation blocks unchanged except block j , where we substitute

$$\theta'_{j,\ell} = \theta_{j,\ell} + z_{a(\pi_{jr}(\ell))} \pmod{1}, \quad \ell \in M_j. \quad (19)$$

Here and below a phase used as a real rotation is represented in $[0, 1)$. For each fixed z , this substitution preserves the product uniform distribution of the rotation inputs. Even when several coordinates receive the same shift, each coordinate undergoes a fixed translation, so their conditional joint law remains the same product law.

To compare the two function evaluations, let $y \in \mathbb{T}^{M_c}$ be the phase vector in $h_{I, t|_I}(\theta')$. For every $\kappa \in M_b$, composition of the projections gives

$$y_{\pi_{bc}(\kappa)} = z_{a(\pi_{br}(\kappa))} + \sum_{k \in B} t_k \theta_{k, \pi_{bk}(\kappa)} \pmod{1}. \quad (20)$$

Indeed, the omitted i term is zero, and the changed j term has coefficient exactly one. This last fact is essential: reducing the shifted rotation modulo one before multiplying would not in general be valid at a fractional coefficient. Thus $H(t, \theta; z) = F_b(y \circ \pi_{bc})$. If $c = b$ the two evaluations are identical; if deleting i changes the minimum layer, the compatibility bound (7) still gives

$$|H(t, \theta; z) - h_{I, t|_I}(\theta')| \leq 2L/D < \gamma. \quad (21)$$

Let $g_{I,t|_I}$ be the fixed junta approximating $h_{I,t|_I}$ in L^2 norm to error less than γ . Its selected coordinates in block k belong to S_I^k . Product-law preservation for every fixed z implies

$$\|h_{I,t|_I}(\theta') - g_{I,t|_I}(\theta')\|_{L^2(\theta,z)} < \gamma.$$

Moreover, the composed function $g_{I,t|_I}(\theta')$ is pointwise independent of z_i . Only the modified block j introduces any z dependence. Every selected coordinate $\ell \in S_I^j$ satisfies $\pi_{jr}(\ell) \in A_I$ by (11), and alignment therefore gives $a(\pi_{jr}(\ell)) \in I$. Combining this observation with (21) and applying the contraction Q_i on the full product space yields

$$\mathbb{E}_\theta D_i(t, \theta)^2 = \|Q_i H\|_{L^2(\theta,z)}^2 \leq \|H - g_{I,t|_I}(\theta')\|_{L^2(\theta,z)}^2 < (2\gamma)^2. \quad (22)$$

This estimate holds for every fixed $t \in \mathcal{F}$ with $t_i = 0$. Markov's inequality, followed by a union bound over the s indices, therefore gives

$$\Pr_{t,\theta}(t \in \mathcal{F}, \exists i \in B : t_i = 0, D_i(t, \theta) \geq \beta) \leq \frac{s(2\gamma)^2}{\beta^2} < \varepsilon. \quad (23)$$

Positive coefficients. We next bound large influences at positive coefficient levels. This step does not assume that a full coefficient is present. Fix i , all rotations θ , and all coefficients other than t_i . For $0 \leq k \leq m$, let H_k denote (15) with $t_i = k/m$, and put $d_k = \|Q_i H_k\|_2$. Changing one coefficient by $1/m$ changes each phase coordinate by circle distance at most $1/m$, since the rotations lie in $[0, 1)$. The norm-one property of Q_i gives

$$|d_k - d_{k-1}| \leq \|Q_i(H_k - H_{k-1})\|_2 \leq L/m < \beta/2 \quad (1 \leq k \leq m). \quad (24)$$

Write $p_k = \lambda^k / \sum_{h=0}^m \lambda^h$ for the probability of coefficient level k/m . Since $p_k = \lambda p_{k-1}$, (24) implies, for the fixed background data,

$$\begin{aligned} \sum_{k=1}^m p_k \mathbf{1}_{\{d_k \geq \beta\}} &\leq \lambda \sum_{h=0}^{m-1} p_h \mathbf{1}_{\{d_h \geq \beta/2\}} \\ &\leq \lambda \sum_{h=0}^m p_h \mathbf{1}_{\{d_h \geq \beta/2\}}. \end{aligned}$$

Integrating over the fixed data and summing over $i \in B$, we obtain from (18)

$$\begin{aligned} \Pr_{t,\theta}(\exists i \in B : t_i > 0, D_i(t, \theta) \geq \beta) &\leq \lambda \sum_{i \in B} \Pr_{t,\theta}(D_i(t, \theta) \geq \beta/2) \\ &\leq \lambda K < \varepsilon. \end{aligned} \quad (25)$$

Decreasing a level can destroy the only full coefficient. This causes no restriction here: (18) holds for every coefficient vector, including all images of the one-level decrease.

From influences to mean square. The parameter choices in (14) give $\Pr(t \notin \mathcal{F}) < \varepsilon$. By (17), the event $X(t, \theta) > \varepsilon$ is contained in the union of this event and the two events bounded in (23) and (25). Consequently

$$\Pr_{t,\theta}(X(t, \theta) > \varepsilon) < 3\varepsilon.$$

Since $0 \leq X \leq 1$, it follows that

$$\mathbb{E}_{t,\theta,z} H(t, \theta; z)^2 = \mathbb{E}_{t,\theta} X(t, \theta) \leq \varepsilon + \Pr(X > \varepsilon) < 4\varepsilon. \quad (26)$$

We now remove the auxiliary phases and return to the actual grid sampling. This is the only remaining passage from the analytic test to the conditional vertex density.

Removing the phases and discretizing. Define the continuous location

$$x(t, \theta) = \left(\sum_{j \in B} t_j \theta_{j, \pi_{bj}(\kappa)} \pmod{1} \right)_{\kappa \in M_b}.$$

For every fixed $t \in \mathcal{F}$, choose an index j with $t_j = 1$. Applying the translation (19) to this block absorbs every z term in (15). For each fixed z the translated rotation array again has exactly its original product law. Hence

$$\mathbb{E}_{\theta, z} H(t, \theta; z)^2 = \mathbb{E}_{\theta} F_b(x(t, \theta))^2 \quad (t \in \mathcal{F}). \quad (27)$$

The choice of the full block may depend on t , because this equality is asserted separately for every fixed coefficient vector. In particular, (26) implies

$$\mathbb{E}_{t, \theta} [\mathbf{1}_{\mathcal{F}}(t) F_b(x(t, \theta))^2] < 4\varepsilon. \quad (28)$$

Couple the discrete rotations to the continuous ones by

$$\hat{\theta}_{j, \ell} = \frac{\lfloor P\theta_{j, \ell} \rfloor}{P}.$$

They have the required independent uniform grid law. Each input coordinate changes by less than $1/P$, so the sup circle distance between $x(t, \theta)$ and $x(t, \hat{\theta})$ is at most s/P . The Lipschitz bound and the range $[-1, 1]$ therefore imply

$$|F_b(x(t, \theta))^2 - F_b(x(t, \hat{\theta}))^2| \leq 2Ls/P < \varepsilon. \quad (29)$$

Combining (28), (29), and $\Pr(t \notin \mathcal{F}) < \varepsilon$ gives

$$\mathbb{E}_{t, \hat{\theta}} F_b(x(t, \hat{\theta}))^2 < 4\varepsilon + \varepsilon + \varepsilon = 6\varepsilon. \quad (30)$$

The discrete location belongs to the grid of denominator $D = mP$, so F_b equals the original grid function f there. That function equals one at every location of a vertex of $\mathcal{A}$. Thus the indicator that the sampled vertex belongs to $\mathcal{A}$ is pointwise bounded by $F_b(x(t, \hat{\theta}))^2$. This remains true when several distinct vertices have the same location. Inequality (30) proves the proposition. $\square$

7 Choosing the constants and completing the reduction

The construction and its analysis have identified all the required properties of the fixed parameters. We now choose them in an order that makes the graph reduction possible. The key point is that the list bound d is fixed before the layer count and the source soundness, so it cannot depend on the Label Cover alphabets.

Fix a rational $0 < \delta < 1/3$ and a rational $0 < \varepsilon < \delta/8$. Set $L = 64$ and take β, K from Lemma 2.3 for L, ε . Choose a positive integer m with $L/m < \beta/2$, then a rational $0 < \lambda < 1$ with $\lambda K < \varepsilon$. The coefficient law (1) is now fixed. Since its probability p_m of a full coefficient is positive, choose an integer $s \geq 2$ with $(1 - p_m)^s < \varepsilon$.

Next choose $\gamma > 0$ sufficiently small that $s(2\gamma)^2/\beta^2 < \varepsilon$, and then choose an even positive integer P sufficiently large that, with $D = mP$,

$$\frac{2L}{D} < \gamma, \quad \frac{2Ls}{P} < \varepsilon.$$

This establishes every inequality in (14). Lemma 4.1 gives the alphabet-independent bound

$$d = \max\{1, J(sL, \gamma)(m+1)^s\}.$$

Apply Lemma 5.1 to $s, d, \eta = \varepsilon$, obtaining $r \geq s$ and a rational law μ on $\binom{[r]}{s}$. Finally choose a rational $\sigma \in (0, 1)$ with

$$4^r d^2 \sigma < \varepsilon. \quad (31)$$

Only now invoke Theorem 2.1 for this fixed σ . Its constant alphabets may be large, but no preceding choice depends on their sizes or on the input formula.

All objects used to run the reduction are finite integers or finite rational probability tables. For each fixed δ , choose them once and incorporate their finite descriptions into one algorithm. Neither a uniform runtime as $\delta \rightarrow 0$ nor an algorithm computing all constants from δ is asserted or needed. There is no input-dependent advice. The real tolerances, continuous functions, junta approximants, and alignments enter only the analysis.

Proof of Theorem 1.1. It suffices to treat rational $0 < \delta < 1/3$: for a fixed real threshold, choose a fixed rational $0 < \delta_0 < \delta$ and use the reduction for δ_0 . Fix rational δ and choose the constants as above, with $8\varepsilon < \delta$. Use the preprocessing and graph construction of Section 3.

The two preprocessing branches already have the required promises. A formula with no remaining clauses is satisfiable and produces a one-vertex graph. An empty clause certifies unsatisfiability and produces K_q , where $q > 1/\delta$ and hence $1/q < \delta$. For every other input, Proposition 3.1 gives a deterministic polynomial-time construction of a nonempty finite simple unweighted graph. If the formula is satisfiable, its Label Cover instance has value one, and Lemma 3.2 supplies a proper three-coloring of every output vertex.

Suppose instead that the formula is unsatisfiable, so its Label Cover value is at most σ . If the metric clique branch occurs, the output again has independent-set density $1/q < \delta$. Otherwise, let $\mathcal{A}$ be any nonempty independent set of the output graph. Lemma 5.3 and (31) bound the probability of an unaligned chain- B pair by 2ε . At every aligned pair, Proposition 6.1 bounds the conditional probability of sampling $\mathcal{A}$ by 6ε ; at other pairs it is at most one. The multiplicity expansion in Proposition 3.1 makes this sampling law exactly the uniform law on output vertices. Hence

$$\frac{|\mathcal{A}|}{|V(G)|} \leq 6\varepsilon + 2\varepsilon = 8\varepsilon < \delta.$$

The empty independent set satisfies the same strict bound because the graph is nonempty. Maximizing over all independent sets proves the required soundness promise. $\square$

8 Proof of distributional alignment

We prove Lemma 5.1, the combinatorial input used to choose the sampling law on layer sets. The argument depends only on the list bound and the order of the indices; it has no graph-theoretic or analytic hypotheses.

Proof of Lemma 5.1. Call B *bad* if no map in (13) exists. For each label x that appears on a list indexed inside B , consider

$$\bigcap \{I : \emptyset \neq I \subseteq B, x \in A_I\}.$$

The set B is bad exactly when one of these intersections is empty. Indeed, a nonempty intersection permits a choice of $a(x)$ independently for each appearing label; labels absent from all these lists may be sent to any element of the nonempty set B .

We first prove that for every $\xi > 0$, some $r \geq s$ admits a real probability distribution on $\binom{[r]}{s}$ under which every list family satisfying (12) has bad-set probability at most ξ . The proof has three steps. Finite minimax turns a failure of this assertion into random list patterns making every s -set bad with positive probability. Ramsey homogenization produces an order-invariant limiting pattern on the rationals. Finally, an invariant-cut argument shows that such a limiting pattern has no bad sets.

Finite patterns and minimax. Order each list arbitrarily. On a finite ordered index set, retain only the length of each list and the equality relation among its occupied slots. Each list has at most d slots, and no two occupied slots of one list are equal. There are finitely many resulting patterns. Every actual list family gives such a pattern, and conversely a pattern is realized by using its equivalence classes as labels. Thus the original label universe has no further role in either separation or badness.

Write $\mathcal{C}_r$ for the finite set of patterns on $[r]$ satisfying (12), and put

$$b(B, C) = \mathbf{1}\{B \text{ is bad in } C\}, \quad B \in \binom{[r]}{s}, \quad C \in \mathcal{C}_r.$$

If the assertion with tolerance ξ were false for every r , then for each $r \geq s$ the finite minimax theorem [26] would give

$$\min_{\nu} \max_{C \in \mathcal{C}_r} \mathbb{E}_{B \sim \nu} b(B, C) = \max_{\rho} \min_{B \in \binom{[r]}{s}} \mathbb{E}_{C \sim \rho} b(B, C) > \xi.$$

Here ν and ρ range over the two finite probability simplices; their compactness ensures that the extrema are attained. Consequently, for every $r \geq s$ there would be a random allowed pattern C_r such that

$$\Pr(B \text{ is bad in } C_r) \geq \xi \quad \text{for every } B \in \binom{[r]}{s}. \quad (32)$$

Assume this counterstatement for the remainder of the argument.

An order-invariant limit. The homogenization step applies finite Ramsey theory [23] to discretized probability patterns, following the kind of extraction used by Trotter and Winkler [25]. We give the particular argument needed here in full. For $t \geq 1$, let $\mathcal{P}_t$ denote the finite set of all allowed patterns on $[t]$, using lists on subsets of size at most s . Restriction to an ordered subset, followed by its increasing identification with an initial interval of integers, defines a map between the appropriate pattern spaces.

Fix $k \geq s$. For each $1 \leq t \leq k$ and each t -subset J of $[r]$, the restriction of C_r to J has a probability vector in the finite simplex on $\mathcal{P}_t$. Color J by this vector with every entry placed in a bin of width at most $1/k$. The number of colors depends only on s, d, t, k , not on r . Successive applications of the finite Ramsey theorem, with the required intermediate set sizes chosen backwards, give the following when r is sufficiently large: there is a k -element set H_k on which all the colorings, for $1 \leq t \leq k$, are homogeneous. Passing to a subset preserves each homogeneity already obtained. In particular, the induced pattern laws on any two t -subsets of H_k differ by at most $1/k$ in every coordinate.

Let $P_t^{(k)}$ be the law on the first t points of H_k . If $t \leq u \leq k$ and $J \in \binom{[u]}{t}$, restricting $P_u^{(k)}$ to J gives the law on the corresponding t -subset of H_k . Therefore

$$\left| (P_u^{(k)}|_J)(C) - P_t^{(k)}(C) \right| \leq \frac{1}{k} \quad (C \in \mathcal{P}_t). \quad (33)$$

Compactness of each finite simplex and a diagonal subsequence as $k \rightarrow \infty$ produce limiting laws P_t for every t . Taking limits in (33) gives exact consistency under every ordered restriction:

$$P_u|_J = P_t \quad (t \leq u, J \in \binom{[u]}{t}). \quad (34)$$

Moreover, (32) gives

$$P_s(\text{the full index set is bad}) \geq \xi. \quad (35)$$

Assign the law P_t to every increasing t -tuple of rational indices. The consistency in (34) constructs a countable random array of list lengths and equality relations indexed by the nonempty $I \subseteq \mathbb{Q}$ of size at most s . For completeness, enumerate $\mathbb{Q}$ and successively extend the pattern on the first n enumerated indices to the first $n + 1$, using the conditional probabilities supplied by their consistent finite laws. The choices on zero-probability conditioning events are immaterial. Every finite set of rational indices then has its specified law.

All equivalence-relation axioms, distinctness within a list, and separated disjointness hold simultaneously almost surely: each violation involves finitely many indices, and there are only countably many possible violations. The array's law is invariant under every increasing automorphism of $\mathbb{Q}$, because its finite laws depend only on order. Equation (35) says that every fixed rational s -set still has bad probability at least ξ .

We have thus retained the positive bad-set probability while removing all distinctions between index sets having the same order type. We next show that this order invariance, together with separated disjointness, forces every label to have an index common to all its occurrences.

An invariant cut for each label. Fix a nonempty $I \subseteq \mathbb{Q}$, $|I| \leq s$, and one of its d possible slots. Let E be the event that this slot is occupied. On E , write $\mathcal{L}$ for its equivalence class of occupied slots, and define

$$p = \sup\{\min J : \text{a slot at } J \text{ belongs to } \mathcal{L}\}. \quad (36)$$

The set whose supremum is taken contains $\min I$. It is bounded above by $\max I$, since an occurrence at J with $\min J > \max I$ would violate separated disjointness. Thus

$$\min I \leq p \leq \max I \quad \text{on } E. \quad (37)$$

The random variable p is measurable on E : the occurrence family is countable, and testing whether its supremum is at most a given real number is a countable intersection of measurable conditions on slots.

Every increasing automorphism g of $\mathbb{Q}$ extends uniquely to an increasing homeomorphism $\bar{g}$ of $\mathbb{R}$. Transporting the array by g transports the occurrence family in (36) and sends its supremum to $\bar{g}(p)$. If g fixes I pointwise, it also fixes the selected slot and its occupancy event. Hence the finite measure

$$\nu(U) = \Pr(E \text{ and } p \in U), \quad U \subseteq \mathbb{R} \text{ Borel},$$

is invariant under all such $\bar{g}$. This formulation also covers $\Pr(E) = 0$ without conditioning on a null event.

Let $a < b$ be consecutive points of I . For any rational $a < q < q' < b$, there is an increasing automorphism of $\mathbb{Q}$ fixing I pointwise and carrying q to q' . For example, use a piecewise linear increasing bijection of $\mathbb{R}$ that is the identity off $[a, b]$, maps q to q' , and is linear on $[a, q]$ and $[q, b]$. Its rational breakpoints and rational slopes make it a bijection of $\mathbb{Q}$. Invariance gives

$$\nu((-\infty, q]) = \nu((-\infty, q']), \quad \text{so} \quad \nu((q, q']) = 0.$$

Countably many such rational intervals cover (a, b) , whence $\nu((a, b)) = 0$. Together with (37), this proves

$$\Pr(E \text{ and } p \notin I) = 0. \quad (38)$$

If I is a singleton, (38) follows directly from (37).

There are only countably many possible slots, so (38) holds simultaneously at every occupied slot with probability one. Equivalent slots have exactly the same occurrence family and hence the same supremum p . It follows that every equivalence class has a common index in all the sets on whose lists it occurs. In particular, on every finite rational s -set B , every label appearing inside B has a nonempty intersection of its occurrence sets there. Thus no such B is bad, contradicting (35) and $\xi > 0$. This proves the assertion with real probabilities.

Rational probabilities. Choose $0 < \xi < \eta$ and obtain r and a real distribution μ_0 with bad-set probability at most ξ for every allowed pattern. Approximate μ_0 by a rational point μ of the same finite simplex with

$$\sum_{B \in \binom{[r]}{s}} |\mu(B) - \mu_0(B)| < \eta - \xi.$$

Every bad-set indicator takes values in $[0, 1]$, so this changes its expectation by less than $\eta - \xi$, uniformly over all patterns. The distribution μ therefore has the required guarantee. $\square$

Remark 8.1. For rational $\eta > 0$, the finite data in Lemma 5.1 can also be found by a terminating search. Enumerate $r \geq s$, enumerate its finitely many allowed slot patterns, and minimize their maximum bad-set probability by a rational linear program. Search until its optimum is less than η . The proof with a smaller tolerance guarantees termination. No bound on the label universe, or oracle describing it, is needed.

References

- [1] Emile Anand. Coloring 3-colorable graphs with $O(n^{4/23})$ colors via a gaussian-cover recursion. arXiv:2610.01071, 2026. October 1, 2026, version 1, Theorem 1. <https://arxiv.org/abs/2610.01071v1>.
- [2] Sanjeev Arora, Carsten Lund, Rajeev Motwani, Madhu Sudan, and Mario Szegedy. Proof Verification and the Hardness of Approximation Problems. *Journal of the ACM*, 45(3):501–555, May 1998. <https://people.csail.mit.edu/madhu/papers/1992/almss-journ.pdf>.
- [3] Tim Austin. On the failure of concentration for the ℓ_∞ -ball. *Israel Journal of Mathematics*, 211(1):221–238, 2016. Theorem 1.1 in the version arXiv:1309.3315v4, 23 June 2014. <https://arxiv.org/abs/1309.3315v4>.
- [4] Nikhil Bansal, Neng Huang, and Euiwoong Lee. Improved SDP-Based Algorithm for Coloring 3-Colorable Graphs. arXiv:2602.05904, 2026. <https://arxiv.org/abs/2602.05904v1>.
- [5] Libor Barto, Jakub Bulín, Andrei Krokhin, and Jakub Opršal. Algebraic approach to promise constraint satisfaction. *Journal of the ACM*, 68(4):28:1–28:66, 2021. <https://doi.org/10.1145/3457606>.
- [6] Mark Braverman, Subhash Khot, Noam Lifshitz, and Dor Minzer. An invariance principle for the multi-slice, with applications. In *2021 IEEE 62nd Annual Symposium on Foundations of*

- Computer Science*, pages 228–236. IEEE, 2022. Full version, arXiv:2110.10725v2, 25 July 2025; Corollary 1.19. <https://arxiv.org/abs/2110.10725v2>.
- [7] Irit Dinur, Venkatesan Guruswami, Subhash Khot, and Oded Regev. A new multilayered PCP and the hardness of hypergraph vertex cover. *SIAM Journal on Computing*, 34(5):1129–1146, 2005. <https://doi.org/10.1137/S0097539704443057>.
 - [8] Irit Dinur, Subhash Khot, Guy Kindler, Dor Minzer, and Muli Safra. Towards a proof of the 2-to-1 games conjecture? *Theory of Computing*, 21(11):1–50, 2025. <https://theoryofcomputing.org/articles/v021a011/>.
 - [9] Irit Dinur, Subhash Khot, Will Perkins, and Muli Safra. Hardness of finding independent sets in almost 3-colorable graphs. In *Proceedings of the 51st Annual IEEE Symposium on Foundations of Computer Science*, FOCS '10, pages 212–221. IEEE Computer Society, 2010. <https://www.wisdom.weizmann.ac.il/~dinuri/mypapers/DKPS-3col9.pdf>.
 - [10] Irit Dinur, Elchanan Mossel, and Oded Regev. Conditional hardness for approximate coloring. *SIAM Journal on Computing*, 39(3):843–873, 2009. <https://doi.org/10.1137/07068062X>.
 - [11] Irit Dinur and David Steurer. Analytical approach to parallel repetition. In *Proceedings of the 46th Annual ACM Symposium on Theory of Computing*, STOC '14, pages 624–633. Association for Computing Machinery, 2014. Theorem 6.2 in the full version dated 10 June 2014. <https://www.dsteurer.org/paper/productgames.pdf>.
 - [12] Yumou Fei, Dor Minzer, and Shuo Wang. On the hardness of 4-to-1 games with perfect completeness. Electronic Colloquium on Computational Complexity, Report TR26-179, 2026. September 14, 2026. <https://eccc.weizmann.ac.il/report/2026/179/>.
 - [13] Ehud Friedgut. Boolean Functions with Low Average Sensitivity Depend on Few Coordinates. *Combinatorica*, 18(1):27–35, 1998. <https://doi.org/10.1007/PL00009809>.
 - [14] Venkatesan Guruswami and Sanjeev Khanna. On the hardness of 4-coloring a 3-colorable graph. *SIAM Journal on Discrete Mathematics*, 18(1):30–40, 2004. <https://doi.org/10.1137/S0895480100376794>.
 - [15] Venkatesan Guruswami and Sai Sandeep. d -to-1 hardness of coloring 3-colorable graphs with $O(1)$ colors. In *47th International Colloquium on Automata, Languages, and Programming*, volume 168 of *Leibniz International Proceedings in Informatics*, pages 62:1–62:12. Schloss Dagstuhl–Leibniz-Zentrum für Informatik, 2020. <https://drops.dagstuhl.de/entities/document/10.4230/LIPIcs.ICALP.2020.62>.
 - [16] Yahli Hecht, Dor Minzer, and Muli Safra. NP-hardness of almost coloring almost 3-colorable graphs. In *Approximation, Randomization, and Combinatorial Optimization. Algorithms and Techniques*, volume 275 of *Leibniz International Proceedings in Informatics*, pages 51:1–51:12. Schloss Dagstuhl–Leibniz-Zentrum für Informatik, 2023. <https://drops.dagstuhl.de/entities/document/10.4230/LIPIcs.APPROX/RANDOM.2023.51>.
 - [17] Ken-ichi Kawarabayashi, Mikkel Thorup, and Hirotaka Yoneda. Better coloring of 3-colorable graphs. In *Proceedings of the 56th Annual ACM Symposium on Theory of Computing*, pages 331–339. Association for Computing Machinery, 2024. <https://arxiv.org/abs/2406.00357>.

- [18] Sanjeev Khanna, Nathan Linial, and Shmuel Safra. On the hardness of approximating the chromatic number. *Combinatorica*, 20(3):393–415, 2000. <https://doi.org/10.1007/s004930070013>.
- [19] Andrei Krokhin, Jakub Opršal, Marcin Wrochna, and Stanislav Živný. Topology and adjunction in promise constraint satisfaction. *SIAM Journal on Computing*, 52(1):38–79, 2023. <https://doi.org/10.1137/20M1378223>.
- [20] Edward J. McShane. Extension of range of functions. *Bulletin of the American Mathematical Society*, 40(12):837–842, 1934. <https://doi.org/10.1090/S0002-9904-1934-05978-0>.
- [21] Ijay Narang and Yukai Tang. Improved SDP coloring of 3-colorable graphs from recursive gaussian certificates. arXiv:2609.33684, 2026. September 27, 2026, version 1, Theorem 1.1. <https://arxiv.org/abs/2609.33684v1>.
- [22] OpenAI. Perfect completeness for 2-to-1 games. OpenAI Math Release preprint [OAI:Perfect-completeness-for-2-to-1-games-September-23-2026](#), 2026.
- [23] Frank P. Ramsey. On a problem of formal logic. *Proceedings of the London Mathematical Society*, 30:264–286, 1930. <https://doi.org/10.1112/plms/s2-30.1.264>.
- [24] Ran Raz. A Parallel Repetition Theorem. *SIAM Journal on Computing*, 27(3):763–803, 1998. <https://doi.org/10.1137/S0097539795280895>.
- [25] William T. Trotter and Peter Winkler. Ramsey theory and sequences of random variables. *Combinatorics, Probability and Computing*, 7:221–238, 1998. <https://doi.org/10.1017/S0963548398003393>.
- [26] John von Neumann. Zur theorie der gesellschaftsspiele. *Mathematische Annalen*, 100:295–320, 1928. <https://doi.org/10.1007/BF01448847>.